

Accessible VR social stories in education for neurodiverse students

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ABSTRACT

Immersive environments have the potential to revolutionize education by enhancing motivation and enriching learning experiences. Recent research suggests that VR-based experiences can be particularly beneficial for neurodiverse students, aiding their psychosocial integration and skill-building. Still, questions related to the accessibility of VR provided solutions in educational contexts remain unsolved. This paper presents the InclusionVRity project which is an EU funded project aiming at providing teachers and caregivers with accessible VR tools and materials for their application in educational contexts and learning activities to address the needs of neurodiverse students.

Index terms: Virtual reality, Accessibility, Social stories, neurodiversity.

1 INTRODUCTION

Accessibility and inclusion are becoming a priority in international policy frameworks for education. The EU sets a high priority for the inclusion and diversity of all societal groups, especially through the Rights of Persons with Disabilities Strategy 2021-2030, and the adoption of the UN Agenda 2030 through the Sustainable Development Goals (SDG). In this regard, SDG 4 - Target 4.5 aims to 'eliminate gender disparities in education and ensure equal access to all levels of education and vocational training for the vulnerable, including persons with disabilities, indigenous peoples and children in vulnerable situations by 2030'.

Neurodiversity impacts how people process information, as it can affect people's perception, memory, language, attention, problem solving, and comprehension. Terminology for categories and conditions can be related to cognitive and learning disabilities, and include intellectual disabilities, developmental disabilities; attention deficit hyperactivity disorder (ADHD), autism, dementia, dyslexia, and more (Web Accessibility Initiative¹). If the different needs of all students are not considered when developing tools and materials, neurodiverse students might be exposed to barriers affecting their education and social life, like vulnerability to bullying or prone to the risk of early school-leaving and low proficiency in basic skills to name a few [1,2,6].

2 VR SOCIAL STORIES IN EDUCATION

Recent research studies have proved that VR can be an ideal tool for intervention and rehabilitation for neurodiverse students by providing them with a real-life scenario, in a more friendly environment. What is even more appealing about VR is that the experience can be dynamic, individualized, and personalized for the user's preferences [9]. Therefore, interaction with VR technology as a method for learning may contribute to improving

their learning experience. Yet, a key challenge is that most educational VR simulations are not designed as a tool to be used at different educational levels and with different pedagogical approaches, but as stand-alone learning experiences [2]. For VR to become a relevant tool for instructors, these professionals should be able to create and edit their own content. This is starting to happen with content based on 360° videos, which can form the basis of subsequent educational activities [2,3]. The merging of VR with social stories, describing everyday situation, can have numerous benefits for neurodiverse students, including the overall improvement of their social skills. Recent research [2,3,4,] has also identified that the transformation of social stories into VR tools can be constructive in the gamified training of neurodiverse students in terms of their psychosocial integration and capacity building. However, it is crucial to critically consider accessibility from the design phase. One of the major barriers for the use of VR technologies in educational contexts is the lack of accessible content, along with the lack of technical skills, training and resources [4,5].

3 ACCESSIBILITY IN VR

Immersive environments, such as VR and AR present new transformative opportunities around how we work, communicate, collaborate, and engage with others. However, research has highlighted significant accessibility barriers associated with these technologies that can lead to the exclusion of disabled people [7].

In the case of VR, main key accessibility challenges for the use of this technology with neurodiverse users include: sensory and information overload, cybersickness, discomfort from head-mounted displays (HMDs), physical or mental stress, and inadequate support and training [4]. For people with autism spectrum disorder (ASD), research has identified barriers related to the lack of customization, hypersensitivity to sounds, difficulties with interpersonal spaces, and vestibular issues [8]. For students with dyslexia research has identified difficulties related to phoneme decoding, visuospatial attention, and semantic development [9].

In all studies questions related to accessibility, affordability, customization, and cost were reported as main drawbacks to consider when implementing VR materials and tools in educational contexts. In this regard, even if international legislative frameworks on accessibility are being implemented world-wide, specific standards and guidelines addressing accessibility in immersive environments remain scarce. Existing accessibility recommendations in immersive environments include XAUR², (which is also linked to WCAG2.1, updated to WCAG2.2³), and work being conducted at ITU [6]. Still, these recommendations remain unknown or cumbersome to most VR developers and experts.

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²XR Accessibility User Requirements <https://www.w3.org/TR/xaur/>

³Web Content Accessibility Guidelines (WCAG2.2).
<https://www.w3.org/TR/WCAG22/>

4 INCLUSiVRity PROJECT

InclusiVRity is an EU funded project which aims at empowering teachers with VR technologies to foster a more inclusive and accessible environment for neurodiverse students. The main objective of the project is to help teaching professionals and caregivers of neurodiverse students to include innovative VR tools and materials for the development of key skills, such as social, communication, teamwork, and problem-solving.

To address these needs the project will use a VR platform for the implementation of specific scenarios with social stories, which are short narratives that describe everyday life situations using paper or video [4]. These scenarios will be co-created between teachers/caregivers and students to address specific needs related to neurodiversity. The criteria to develop the co-created scenarios in the VR platform will be: i) to simulate the real-world school/social/life situations (i.e., what do the users experience?); ii) to address common barriers that neurodiverse students face; iii) to enhance social, educational and functional skills (e.g., communication, reading, focusing) to facilitate the users' independence, access and confidence in life/social activities; iv) to raise awareness about neurodiversity among all students. This intersectoral approach is key to ensure that the production of VR stories is created and validated from an end user perspective.

Each scenario will consist of one or more interconnected 360° videos enriched with visual interactive elements in the form of hotspots. Students will be able to make their own choices, and visual and auditory feedback will be provided according to their preferences/needs to stimulate memory, collaborative work among students with accessibility in mind, and add engagement to the VR experience. In terms of accessibility considerations, co-created scenarios and social stories will be created from a user-centred perspective following existing accessibility recommendations in immersive environments. In addition, as sensory and information overload has been reported as a major challenge for neurodiverse users, recommendations and standards for Easy Language, which aim to facilitate reading and language comprehension [10] will be considered to address common problems and barriers, such as processing information, attention deficit and focus, social interaction, or memory. Simplified information might be preferred as it causes less distraction and allows for greater focus on the primary visual content [6]. Also, the concept of Easy Audio from written to oral texts in Easy Language will be explored. This concept is currently being researched in the WEL project⁴ for video games and cultural visits.

5 CONCLUSION

It is expected that in the coming years immersive environments, such as VR, become an integral part in education for the provision of more engaging learning experiences. The use of this technology has proved to be effective to improve psychological, functional, educational, social interactions and related skills (communication, reading, focus), the most common barriers neurodiverse students face. While VR environments can present solutions for neurodiverse students, if VR content and tools do not consider accessibility from the design phase, these students might be left behind. The InclusiVRity project aims at providing teachers and caregivers with materials and tools to help them understand the use of VR technologies for the pedagogy of neurodiverse students with accessibility in mind. It will also provide a VR platform with embedded co-created social stories depicting potential real-life scenarios that neurodiverse students may face helping their

psychosocial integration and skill-building and also raising awareness about neurodiversity in educational contexts.

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⁴ The Wel project <https://webs.uab.cat/wel/en/>